

Jiefu Zhou (Jeff)

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TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, PySpark, PyTorch, TensorFlow, SHAP, Matplotlib), SQL, R, SAS

Tools: MLflow, AWS, Power BI, SAS EG, Git, Hugging Face, LangChain, ChromaDB

Certifications: Microsoft Power BI Data Analyst Associate (PL-300), Google Advanced Data Analytics Professional Certificate

Techniques: Machine Learning, Deep Learning, LLMs, RAG, Causal Inference & A/B Testing, NLP, Prompt Engineering, Feature Engineering, Model Validation, Financial Risk Modeling, Statistical Modeling, Monte Carlo Simulation, Data Visualization

EDUCATION

Rotman School of Management, University of Toronto

Toronto, Canada

Master of Management Analytics (MMA)

06/2026

University of Ottawa

Ottawa, Canada

Honours Bachelor of Commerce (Finance)

04/2025

PROFESSIONAL EXPERIENCE

Bank of Montreal (BMO) — Contract, Full-time

Toronto, Canada

AI Analyst & Enablement, Audit Data Science

Aug 2026 – Feb 2027

- Embed natural language processing (**NLP**), **AI**, and advanced **data analytics** into the corporate audit lifecycle to support AI enablement and automation.
- Built a **Python** ingestion pipeline to parse and normalize heterogeneous structured and **unstructured data** (emails, text files, excel csv) into a unified corpus, partially replacing manual evidence consolidation.
- Designed a **RAG pipeline** with format-aware **chunking**, hybrid **embeddings**, and retrieval strategy, combined with **prompt engineering** to ground **LLM** outputs in traceable source evidence.

Bank of Montreal (BMO)

Toronto, Canada

Data Analyst, Marketing Advanced Data Science

Jan 2026 – Jun 2026

- Wrangled large-scale, messy datasets with missing values, inconsistent formatting, and data-quality issues, building robust preprocessing pipelines to ensure model-ready inputs.
- Engineered scalable data pipelines using **SQL/Python** to integrate customer profile, account, and household data, enabling reproducible model training, validation, and deployment-ready analytics workflows.
- Developed explainable machine learning models (**LightGBM/XGBoost/CatBoost**) to predict 12-month customer lifetime value (CLV) on a **6M+ customer database**, supporting value-based customer prioritization and campaign targeting.
- Built customer segmentation models (**K-Means/GMM/K-Prototypes**) to identify high-value personas and improve precision targeting for marketing campaigns.
- Translated technical outputs into actionable business insights, conducted comprehensive data visualization in **Python/Power BI**.
- Designed randomized controlled trial (**RCT**) methodologies to evaluate campaign effectiveness, measure incremental lift, and validate model-driven targeting recommendations.

Zevnodata Fintech

Toronto, Canada

Business Analyst Intern

May 2023 – Aug 2023

- Cleaned and processed Lululemon's transaction dataset (~200k rows) in IBM SPSS Modeler, applying **C5 decision trees** and K-Nearest Neighbor (**KNN**) to improve model balance and achieve **6% higher prediction accuracy** on growth forecasts.
- Consolidated and analyzed 50k+ external records from Capital IQ, Statista, and internal company reports, producing a centralized dataset that accelerated business performance analysis.
- Performed data analysis and visualization (Matplotlib, Seaborn) in Python, uncovering patterns that surfaced revenue-growth and expansion opportunities and informed strategic market-entry recommendations presented to client stakeholders.

TECHNICAL PROJECTS

- **Generative AI — LLM Agentic Application:** Developed a conversational support agent for international students seeking global work-experience guidance, integrating retrieval-augmented generation (RAG), multi-step reasoning, tool use, and safety guardrails to deliver grounded responses and reduce hallucinations.
- **Fraud Detection — AML Crime Lab, Scotiabank x SAS:** Built an end-to-end, account-level pipeline for detecting credit-card fraud and credit abuse on SAS Viya, combining machine learning models with a rule-based engine for both precision and interpretability, achieving a 90.4% fraud capture rate at 43% of the manual review workload.
- **Natural Language Processing — NLP Sentiment Classification:** Benchmarked traditional ML (TF-IDF with Logistic Regression / Naive Bayes), a Bi-LSTM, and a fine-tuned DistilBERT transformer on 100K+ Amazon reviews to classify binary sentiment.

ADDITIONAL

- Ranked top 8% worldwide in rapid games on Chess.com.